

1. Connector & input protection

Field connector, 2-stage bidirectional
TVS surge protection, chassis
isolation

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: HMT500(260313A)_connector.kicad_sch
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2. Power

eFuse (reverse/OV/UV), 60V buck to 5V,
LDO 3.3V

: HMT500(260313A)_power.kicad_sch

3. MCU

STM32G0B1CCT6, SWD, reset, internal
temperature sensor

: HMT500(260313A)_mcu.kicad_sch

4. Measurement

Capacitive humidity sensor (PCAP04)
and Pt1000 2-wire (ADS1220)

: HMT500(260313A)_measurement.kicad_sch

5. Analog outputs

2x DAC8760 V/I output with TPS26611
miswiring protection

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: HMT500(260313A)_analog_out.kicad_sch
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6. RS-485

THVD2410 +/-70V fault-protected
transceiver

: HMT500(260313A)_rs485.kicad_sch

7. Aux

AO supply VAO 16.1 V (buck), supply
monitor

: HMT500(260313A)_aux.kicad_sch

DOTECH HMT500(260313) - Oil Moisture Transmitter - Schematic v0.9 - Project No. HMT500(260313A)

File names (schematic, PCB, Gerber) = HMT500(260313A).* PCB silkscreen marking = HMT500(260313A)

Signal flow: J1 field connector -> protection -> eFuse -> 5 V buck -> 3.3 V LDO; sensor head J3 -> PCAP04 / ADS1220 -> MCU -> DAC8760 x2 / THVD2410 -> J1

Outputs: RS-485 Modbus RTU + 2x analog (4-20 mA / 0-20 mA / 0-10 V / 0-5 V selectable). Supply 12-28 V DC.

v0.10: AO D30/D40 TVS1401 +/-14 V. AO +/-28/30 V continuous miswiring is NOT guaranteed (#41).

Power symbols: GND, +3V3, +3V3A (analog 3.3 V), +5V, VIN_P (protected input), CHASSIS. Inter-sheet signals: global labels.

History v0.3-v0.8 (MCU G0B1, JST harness, DAC8760/TPS26611 datasheet pinouts, RTN isolation, gated DAC SCLK ...): docs/decision-log.md #9-#26.

v0.9: 3-pass review (docs/hw/final-review-260313A.md): OVP 32.6 V (R5 36.5k); AO supply VAO 16.1 V buck U15 (DAC AVDD, IPS26611 +Vs; dec. #37, IVS320 AO); D3 series Schottky; RS-485 TVS D60/D61; D2 -> TVS3301; eFuse IN C2; buck CIN;

RS485_DE 10k pull-down (in-house bootloader, dec. #31); THVD2410; X7R only; R2 HV 1206; SGOOD -> PA0/PA4; CS_CDC -> PA15; VIN_SENSE PB0;

LATCH/CS pull-ups; OPA197 V+ = DAC AVDD; R31/R41 100k; SWD no-connect flags removed; sheet Aux: +Vs clamp, VIN monitor (DAC boost / HSE / sensor ESD: no room, dec. #32).

Footprints: KiCad 7.0.11 library + project library HMT500_260313A (GDT Bourns 2035-xx-SM, TVS3301 pad floating).

표기 규칙: 부품번호 = 굵은 글자 (R1, U4) / 부품값 = 보통 글자 (10k, DAC8760) / 전원 네트 = 기울인 글자 (+3V3, VIN_P)

신호 네트 = 테두리 있는 라벨 (SPI_SCK, OUT1_EXT) - 네트 이름은 부품번호·부품명과 겹치지 않게 지음

라벨 모양 = 신호 방향: 뾰족한 쪽이 밖 = 이 시트에서 내보냄(출력), 안 = 받음(입력), 양쪽 = 양방향, 네모 = 아날로그·수동

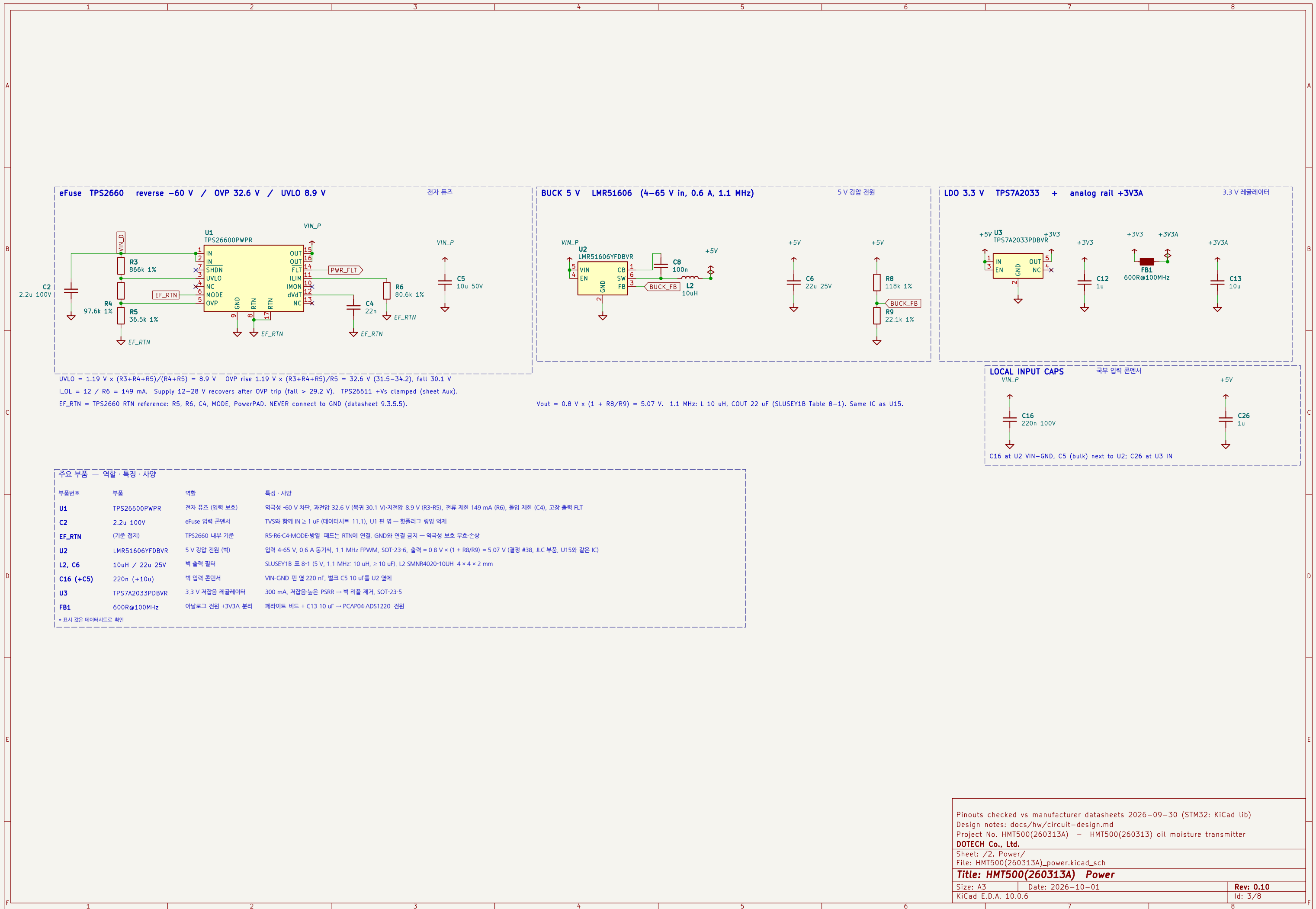
글자 겹침: 생성기가 부품 몸체·핀·선·라벨·메모와 겹치지 않는 자리에 자동 배치하고, 겹치면 생성 실패로 처리

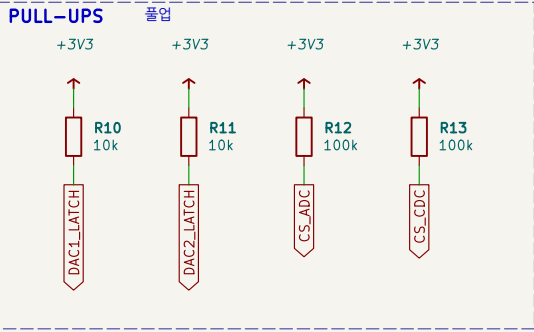
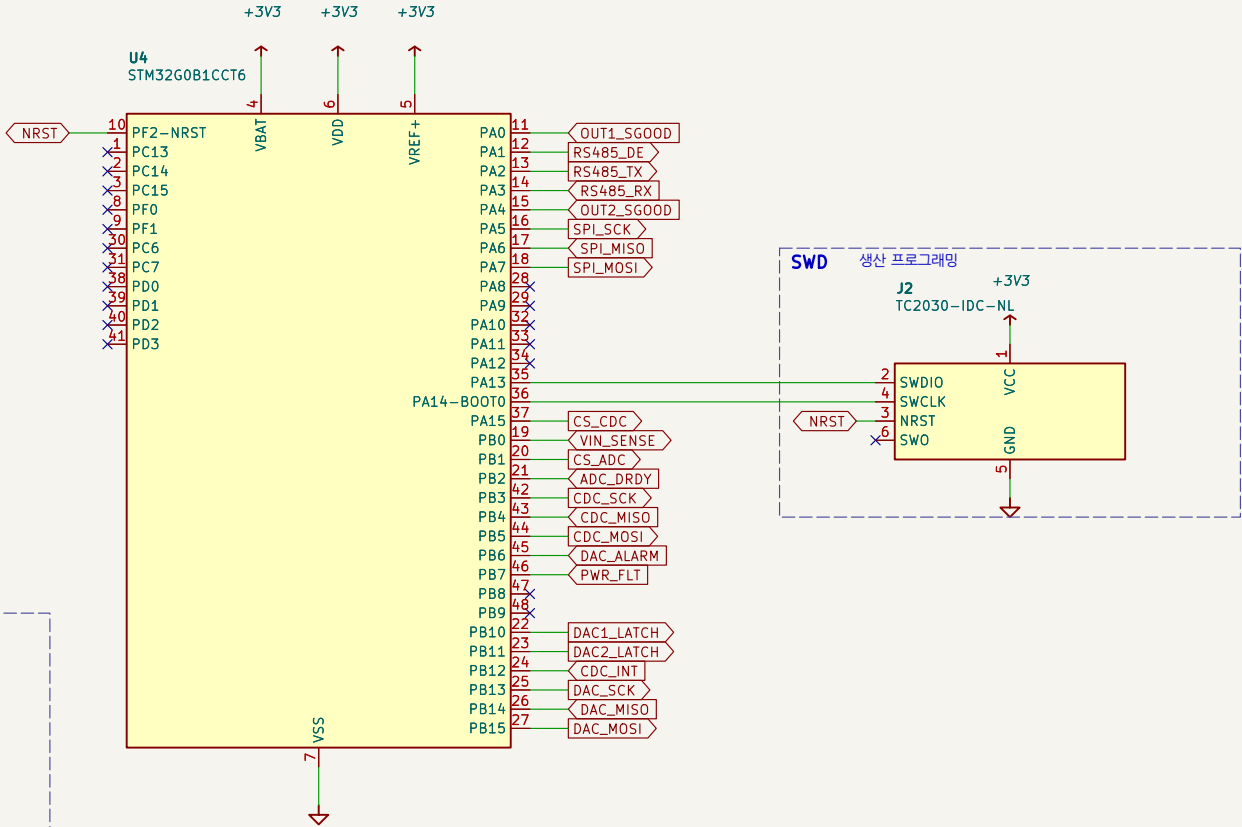
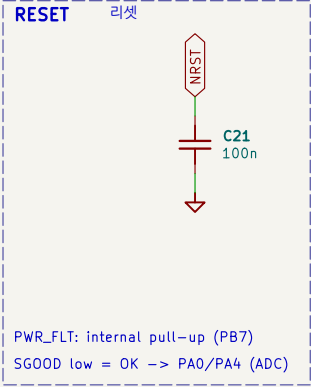
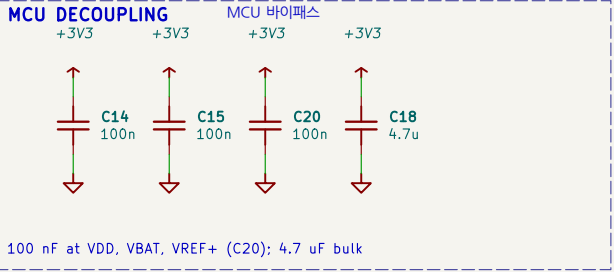
Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)
Design notes: docs/hw/circuit-design.md
Project No. HMT500(260313A) - HMT500(260313) oil moisture transmitter
DOTEC Co., Ltd.

Sheet: /
File: HMT500(260313A).kicad_sch

Title: HMT500(260313A) Top

Size: A3	Date: 2026-10-01	Rev: 0.10
KiCad E.D.A. 10.0.6		Id: 1/8





MCU NOTES (STM32G0B1) MCU 설정 메모

BOOT0 shares PA14/SWCLK - no pull-down.

Option bytes: nBOOT_SEL=1, nBOOT0=1, IWDG_SW=0, RDP 1

-> in-house RS-485 bootloader 0x08000000 (dec. #31).

VDDA = VDD (LQFP48); VREF+ tied to +3V3 (C20).

PCB temp (mid): internal sensor, ADC ch TS + TS_CAL.

Cortex-M0+: no SWO (J2 pin 6 NC).

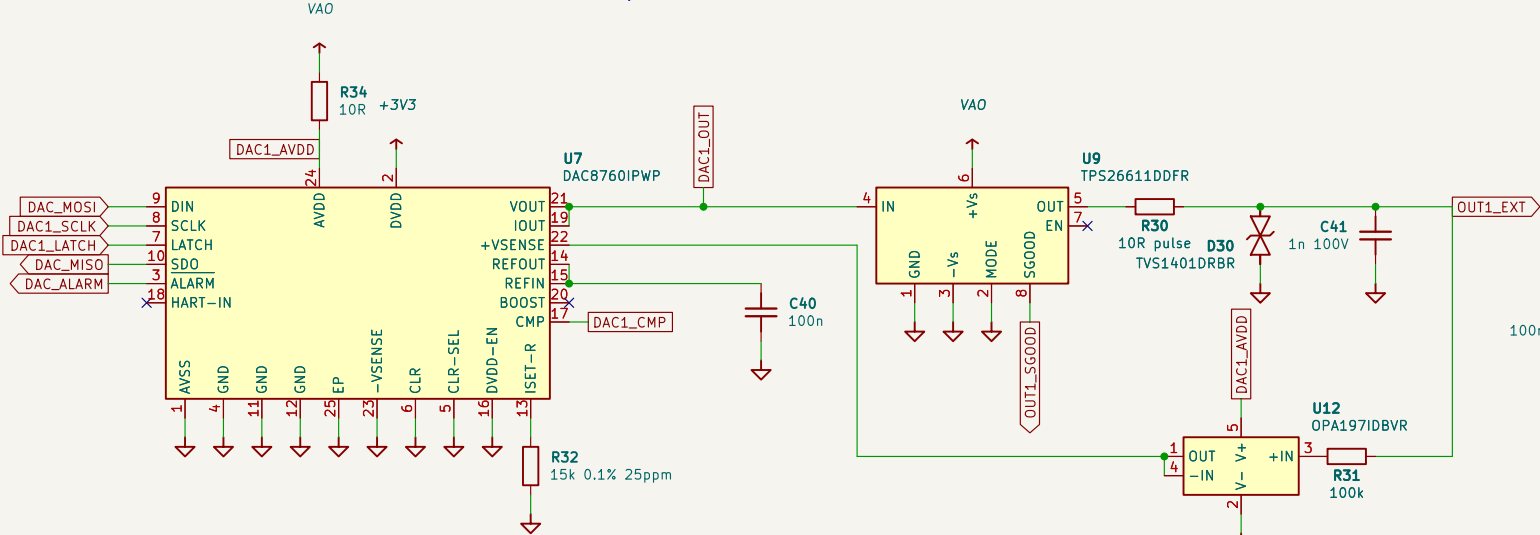
Same die as DP2000 (G0B1CCT6); T3 = -40..125 C.

VIN_SENSE (PB0 ADC_IN8) = VIN_P / 11.

주요 부품 — 역할 · 특징 · 사양			
부품번호	부품	역할	특징 · 사양
U4	STM32G0B1CCT6	계어통신·보정 연산	Cortex-M0+ 64 MHz, 플래시 256 KB, RAM 144 KB, -40~85 °C (샘플 #39), LQFP48, 내부 온도센서 (PCB 온도), SPI1 = ADS1220, SPI2 = DAC (모드 3, SCLK 게이트), SPI3 = PCAP04
J2	TC2030-IDC-NL	생산 때 펌웨어 기록 (SWD)	PCB 패드만 (부품 없음). 이후 업데이트는 RS-485 부트로더
C21	100n	리셋 노이즈 필터	NRST 핀, 물딩 후 EMC 여유
R10, R11	10k	DAC LATCH 풀업	리셋 중 LATCH High -> SCLK 게이트 닫힘, DAC 오입력 방지 (v0.9)
R12, R13	100k	CS 풀업	리셋 중 ADS1220-PCAP04 선택 해제 (디지털 입력 부동 금지)

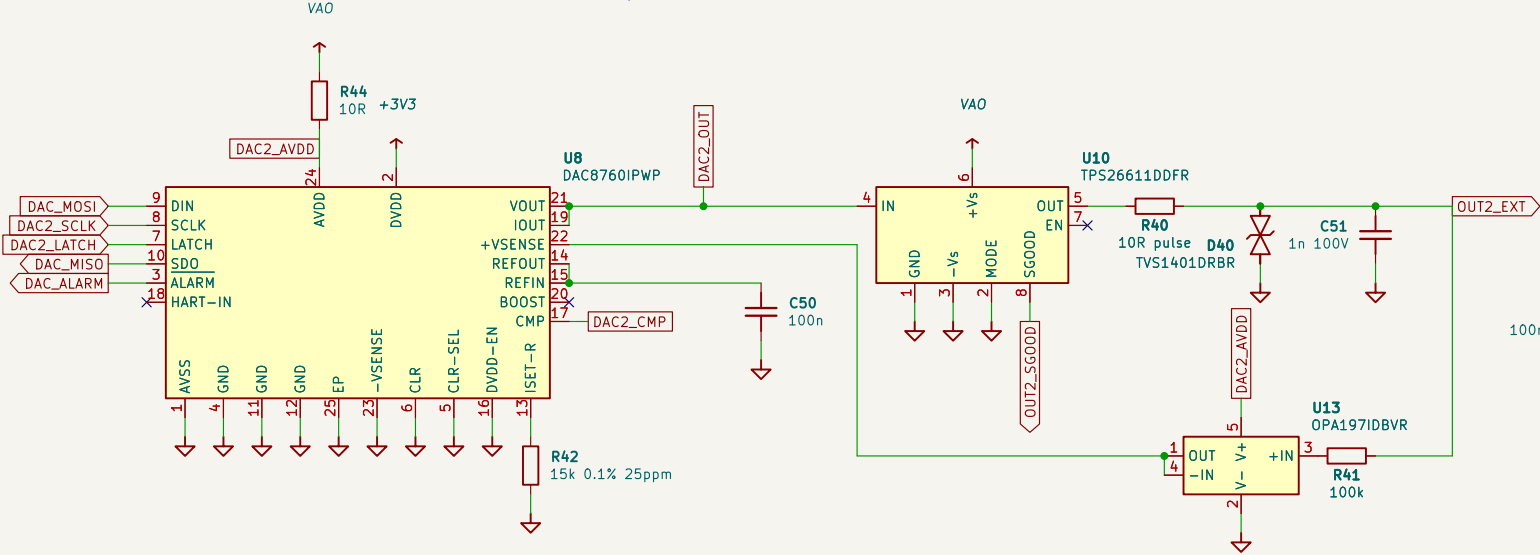
* 표시 값은 데이터시트로 확인

ANALOG OUTPUT CH1 - V/I selectable, TVS +/-14 V -> connector pin 4



VOUT + IOUT one terminal (datasheet 9.1.1.3). +VSENSE = U12 OPA197 follower of terminal OUT1_EXT (after R30) via R31:
no +VSENSE 60k leakage in current mode, series-R drop corrected in voltage mode. ISET-R = R32 15k 0.1%. SGOOD low = OK.

ANALOG OUTPUT CH2 - V/I selectable, TVS +/-14 V -> connector pin 5



VOUT + IOUT one terminal (datasheet 9.1.1.3). +VSENSE = U13 OPA197 follower of terminal OUT2_EXT (after R40) via R41:
no +VSENSE 60k leakage in current mode, series-R drop corrected in voltage mode. ISET-R = R42 15k 0.1%. SGOOD low = OK.

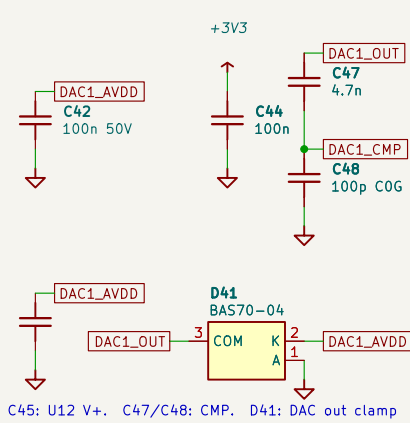
주요 부품 — 역할 · 특징 · 사양

부품번호	부품	역할	특징 · 사양
U7, U8	DAC8760IPWP	아날로그 출력 1·2	16비트, 4-20 / 0-20 / 0-24 mA · 0-5 / 0-10 V 선택, 내부 기준, 고장 알람 — 대체 DAC7760/8750/7750
U9, U10	TPS26611DDFR	출력 오버전 보호	±50 V, 단자 > +Vs 또는 < -0.2 V면 5 us 차단·자동 복귀, 전류 제한 32 mA, RON 7.5 Ω, SGOOD Low = 정상. +Vs = VAO 16.1 V (백, 시트 Aux, 결정 #37)
R30, R40	10R pulse (2512)	서지 전류 제한	출력 보호기와 TVS 사이 직렬
D30, D40	TVS1401DRBR	출력 서지 클램프	±14 V 동작, 클램프 최대 23.55 V @ 30 A/125 C, AO ±28/30 V 지속 오버전 보장 제외 (#41)
C41, C51	1n 100V	출력 고주파 필터	케이블로 들어오는 RF 억제
U12, U13	OPA197IDBVR	+VSENSE 버퍼 (이득 1)	36 V, 입출력 레일투레일 — 단자 전압을 +VSENSE로, 전류 모드 누설 0. V+ = DAC AVDD
R31, R41	100k	버퍼 입력 보호	R30-R40 뒤 단자 전압을 OPA197 + 입력으로 (서지 전류 제한, ±30 V 오버전 9 mW, v0.9)
R32, R42	15k 0.1% 25ppm	외부 ISET-R (전류 기준)	내부 저항보다 온도 드리프트 작음 (REXT 비트 설정)
R34, R44	10R	DAC AVDD 직렬	VAO → AVDD, 전원 상승 속도 제한 (데이터시트 10장)
C47/C48, C57/C58	4.7n / 100p C0G	DAC CMP 보상	VOUT-CMP 4.7 nF + CMP-GND 100 pF (SBA5528D 8.3.2: 470 pF 초과 시 100 pF 추가) — 케이블 용량·보호기 RON 포함 루프 안정 (IVS320 AO rev 1.0)
D41, D51	BAS70-04	DAC 출력 클램프	DAC 출력 줄을 GND - 0.3 V ~ AVDD + 0.3 V로 — TPS26611 차단-재연결 순간 DAC 핀 보호 (외부 검수 H03, IVS320과 동일)
U14	SN74LVC2G32DCUR	DAC별 SCLK 게이트	OR 2회로: DACn_SCLK = SCK OR LATCHn (SPI2 모드 3)
R33	10k	ALARM 플립	오픈 드레인 ALARM 두 개를 한 선으로 (데이터시트: 외부 10 kΩ)

* 표시 같은 데이터시트로 확인

CH1 DECOUPLING

채널 1 바이패스



C45: U12 V+. C47/C48: CMP. D41: DAC out clamp

DAC SELECTION (decision #10)

DAC 선정

U7/U8 fitted: DAC8760 (16-bit, I + V output).

Drop-in alternates (same pinout, no layout change):

DAC7760 12-bit, I + V (cost-down, V/I kept)

DAC8750 16-bit, I only 4-20/0-20/0-24 mA (EE364-equiv.)

DAC7750 12-bit, I only (-5 uA/step vs EE364 2 uA)

Current-only (x750): VOOUT/+VSENSE/-VSENSE absent —

check pin handling before fitting (VERIFY).

DAC SPI — SPI2, SCLK gated per DAC

DAC 전용 SPI

DAC8760 rev D: daisy chain removed; multiple devices

need gated SCLK (datasheet 8.5.1.5, Fig. 8-9).

SPI2 mode 3 (CPOL 1, CPHA 1): SCK idles high, so the

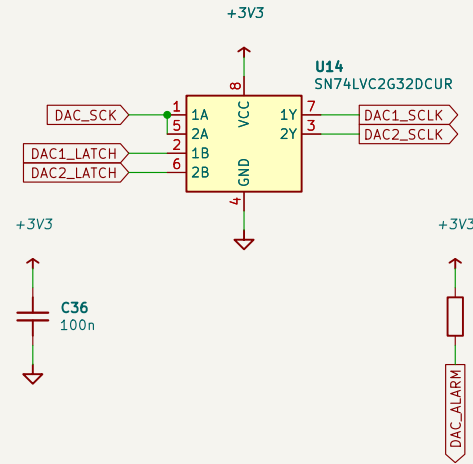
OR gate gives no extra edge at LATCH low/high.

DACn_SCLK = DAC_SCK OR DACn_LATCH (PB10 / PB11).

DIN and SDO shared (SDO 3-state after LATCH).

ALARM: open drain, wired-OR, R33 10k pull-up:

which channel -> read status register.



Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)

Design notes: docs/hw/circuit-design.md

Project No. HMT500(260313A) — HMT500(260313) oil moisture transmitter

DOTECH Co., Ltd.

Sheet: /5. Analog outputs/

File: HMT500(260313A)_analog_out.kicad_sch

Title: HMT500(260313A) Analog outputs

Size: A3

Date: 2026-10-01

Rev: 0.10

KiCad E.D.A. 10.0.6

Id: 6/8

